

综合练习

人员

王承周、许睿谦、阮文璋、王毅博 到课

上周作业检查

上周作业链接: <https://vjudge.net/contest/811695>

Rank	Team	Score	Penalty	A 12 / 17	B 10 / 15	C 5 / 17	D 8 / 14
1	☆ chenxinmiao (陈欣妙)	4	4140	0:31:34	0:59:17	2:12:55:00	6:34:32
2	☆ junyan007008	4	8262	1:11:13 (-1)	11:51:49 (-3)	2:13:24:00 (-3)	2:12:34:58 (-1)
3	☆ lijinsu	4	10746	1:08:15	8:52:45	6:13:05:00	11:20:18 (-2)
4	☆ wcz123bc	4	18706	5:28:47	5:48:28	6:06:09:00 (-6)	6:04:19:56
5	☆ yyycj	4	19376	0:39:33 (-2)	0:55:41	6:15:30:20 (-3)	6:16:10:45
6	☆ wyb1111 (wwyb)	3	1087	5:28:15	5:46:30		6:52:45
7	☆ xuruiqian	3	1185	5:27:03 (-1)	6:02:00 (-2)		7:16:27
8	☆ B_neutrino (Vm1m3)	3	2374	11:55:53	13:57:05		13:41:03
9	☆ r111	2	671	5:25:03	5:46:55		(-3)
10	☆ 7654Fox (隋天乙)	2	20530	7:02:41:32	7:03:29:09		
11	☆ lyc12345bc (刘奕辰)	1	67	1:07:17			
12	☆ chujinxuan (褚锦轩)	1	94	1:14:01 (-1)			

本周作业

<https://vjudge.net/contest/816673> (课上讲了上上周的 A ~ E 题, 课后作业是本周的 A ~ E 题)

课堂表现

今天的 D 题比较抽象一些, 需要同学们课后好好再复盘一下。

课堂内容

CF1512F Education

枚举以每个数作为结尾时需要多少天, 对所有情况求最小值

```
#include <bits/stdc++.h>

using namespace std;

typedef long long LL;
const int maxn = 2e5 + 5;
int a[maxn], b[maxn];

int get_up(int a, int b) { return (a<=0 ? 0 : (a+b-1) / b); }

void solve() {
    int n, c; cin >> n >> c;
    for (int i = 1; i <= n; ++i) cin >> a[i];
    for (int i = 1; i <= n-1; ++i) cin >> b[i];

    LL res = 1e18;
    LL day = 0, sum = 0;
    for (int i = 1; i <= n; ++i) {
        res = min(res, day + get_up(c-sum, a[i])); // 以 i 结尾
        if (i == n) break;

        int k = get_up(b[i]-sum, a[i]); // k 天才能凑够 b[i]
        day += k+1, sum = sum + k*a[i] - b[i];
    }

    // cout << "----- ";
    cout << res << endl;
}

int main()
{
    int T; cin >> T;
    while (T -- ) solve();
    return 0;
}
```

CF1485C Floor and Mod

设 k 为 a/b , 也是 $a\%b$

那么就是 $a = k*b + k$ ($k < b$)

枚举 k , 对应的 b 的范围是 $[k+1, \min(y, (x-k)/k)]$

k 的枚举范围为 $[1, \min(y-1, \sqrt{x})]$

```

#include <bits/stdc++.h>

using namespace std;

typedef long long LL;

void solve() {
    int x, y; cin >> x >> y;
    LL res = 0;
    for (int k = 1; k <= y-1; ++k) {
        int l = k+1, r = min((x-k)/k, y);
        if (r < l) break;
        res += r-l+1;
    }

    // cout << "----- ";
    cout << res << endl;
}

int main()
{
    int T; cin >> T;
    while (T -- ) solve();
    return 0;
}

```

CF1902E Collapsing Strings

构建一棵 trie 存储所有输入的字符串, 然后遍历每个字符串, 翻转这个字符串后在 trie 树中进行查询即可

```

#include <bits/stdc++.h>

using namespace std;

int get_int(char x) { return x - 'a'; }

typedef long long LL;
const int N = 1e6 + 5, M = 26;
int tr[N][M], idx = 0;
int f[N];

void tr_insert(string s) {
    int p = 0;
    for (char i : s) {
        int u = get_int(i);
        if (!tr[p][u]) tr[p][u] = ++idx;
        p = tr[p][u];
        ++f[p];
    }
}

```

```

LL tr_query(string s) {
    int p = 0; LL res = 0;
    for (char i : s) {
        int u = get_int(i);
        if (!tr[p][u]) break;
        p = tr[p][u];
        res += f[p];
    }
    return res;
}

string str[N];

int main()
{
    ios::sync_with_stdio(false);
    cin.tie(0);

    int n, len = 0; cin >> n;
    for (int i = 1; i <= n; ++i) {
        cin >> str[i], len += (int)str[i].size();
        tr_insert(str[i]);
    }

    LL res = 2LL*len*n;
    for (int i = 1; i <= n; ++i) {
        string s = str[i]; reverse(s.begin(), s.end());
        res -= tr_query(s)*2;
    }
    cout << res << endl;
    return 0;
}

```

CF597C Subsequences

$f[i][j]$: 上升子序列的长度为 i , 以 j 这个值结尾的方案数有多少

$f[i][j] <- f[i-1][1] + f[i-1][2] + f[i-1][3] + \dots + f[i-1][j-1]$ (可以用树状数组快速求和)

```

#include <bits/stdc++.h>
#define int long long

using namespace std;

const int N = 1e5 + 5, M = 10 + 5;
int w[N];

int tr[M][N];
int lowbit(int x) { return x & (-x); }
void update(int id, int x, int k) {

```

```

    while (x < N) tr[id][x] += k, x += lowbit(x);
}
int query(int id, int x) {
    int res = 0;
    while (x) res += tr[id][x], x -= lowbit(x);
    return res;
}

signed main()
{
    int n, K; cin >> n >> K; ++K;

    update(0, 1, 1);
    for (int i = 1; i <= n; ++i) {
        int x; cin >> x; x++;
        for (int j = 1; j <= K; ++j) {
            int sum = query(j-1, x-1);
            update(j, x, sum);
        }
    }

    cout << query(K, n+1) << endl;
    return 0;
}

```

CF459E Pashmak and Graph

把所有边按照权值进行分类, 从小到大遍历每种权值的边, 进行 dp

```

#include <bits/stdc++.h>

using namespace std;

const int maxn = 3e5 + 5;
struct node {
    int u, v;
};
vector<node> vec[maxn];
int p[maxn], f[maxn];

int main()
{
    int n, m; cin >> n >> m;
    while (m -- ) {
        int u, v, w; cin >> u >> v >> w; vec[w].push_back({u,v});
    }

    for (int i = 1; i <= 100000; ++i) {
        for (node it : vec[i]) f[it.v] = max(f[it.v], p[it.u]+1);
        for (node it : vec[i]) p[it.v] = f[it.v];
    }
}

```

```
int res = 0;
for (int i = 1; i <= n; ++i) res = max(res, f[i]);
cout << res << endl;
return 0;
}
```